THE END OF TIMETABLE CHAOS

Introducing a Hybrid Intelligence System to revolutionize academic scheduling in Higher Education, aligning with the goals of NEP 2020.

The Scheduling Crisis

Manual timetabling is an inefficient, error-prone process that costs institutions dearly. It leads to frequent class clashes, underutilized resources, and widespread dissatisfaction among both students and faculty.

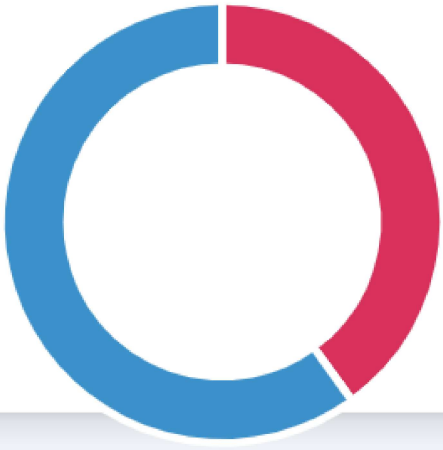
1000s

of Hours Wasted

Administrative staff spend countless hours every semester manually creating and adjusting complex schedules.

Resource Underutilization

Manual methods often lead to inefficient use of classrooms and faculty time, with an estimated 40% of resources being sub-optimally allocated due to clashes and gaps.



15%

Average Error Rate

Manual scheduling frequently results in clashes and errors that require constant, frustrating revisions.

Our Solution: A Smart Hybrid Scheduler

We introduce a novel **Hybrid Intelligence System** that blends algorithmic power with essential human oversight. This is not just an automation tool; it's a new, smarter workflow for academic institutions.

1. Algorithmic Generation

Our custom-built intelligent algorithm processes a complex web of constraints—teacher qualifications, batch

requirements, workload limits, and more—to automatically generate a complete, clash-free timetable with a single click.

2. Human-in-the-Loop Control

A secure, intuitive admin panel provides the crucial final touch. Authorized users can instantly view, verify, and make manual overrides to the schedule, adapting to any real-world exceptions and last-minute changes with ease.

The 5-Step Implementation Flow



Data Management

Admin inputs core data via secure web forms.





One-Click Generation

Admin triggers the intelligent scheduler.





Constraint Solving

Algorithm finds valid, clash-free slots.





Database Update

Timetable is saved to PostgreSQL.





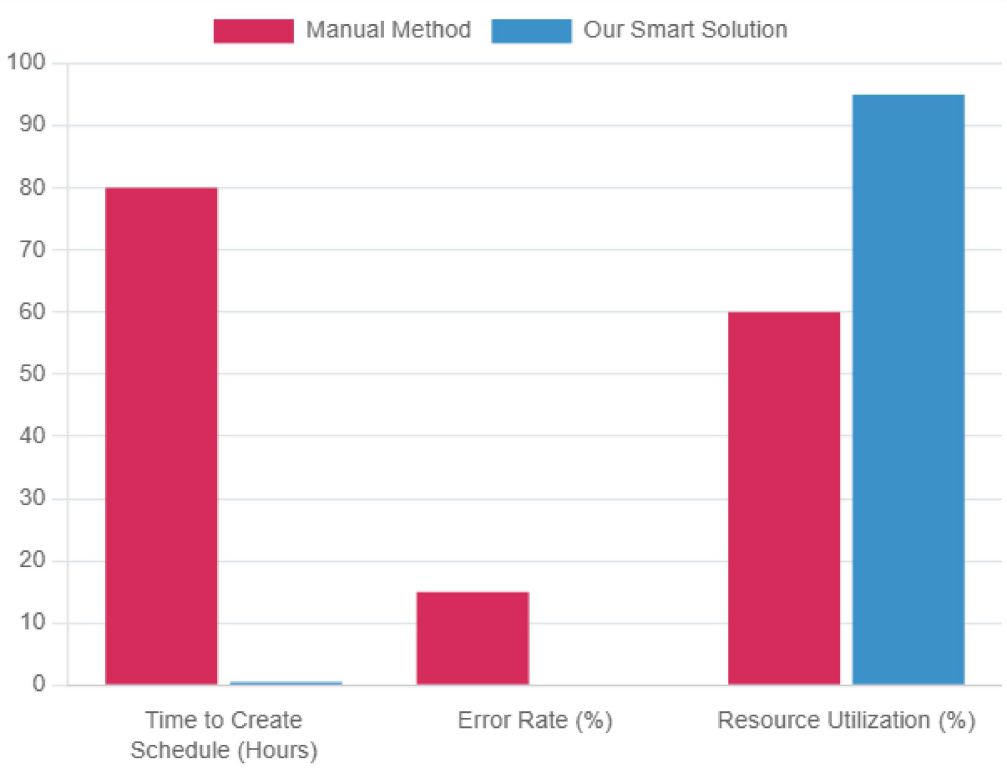
Live Dashboard

Schedule is instantly available to all users.

Quantifiable Impact & Benefits

Manual Method vs. Our Smart Solution

Our platform delivers transformative improvements across key performance indicators, creating a more efficient, accurate, and stress-free academic environment for everyone involved.



Future Scope: The Next Generation of Scheduling

AI-Powered Swap Suggestions

When a teacher takes a last-minute leave, an AI model will suggest the top 3 optimal "swaps" or replacements that cause the least disruption to the existing schedule, providing admins with instant, intelligent solutions.

Predictive Leave Management

Analyze historical data to predict likely faculty leaves based on patterns, allowing the scheduler to proactively build more resilient, "backup-ready" timetables that anticipate potential disruptions.

Student Preference Module

Empower students by allowing them to input preferences for elective courses. The scheduler will then optimize the timetable to maximize student satisfaction based on real, aggregated demand.

Automated Notifications

Integrate with SMS and Email gateways to automatically alert students and faculty of any last-minute changes to their schedule, ensuring everyone is always up-to-date.

Smart Hybrid Timetable Scheduler - A Smart India Hackathon Project